University of Information Technology and Sciences (UITS)
Department of Computer Science and Engineering
Course Title: Software Engineering and System Analysis Lab
Course Code: CSE 0613226
Section: 6B
Course Teacher: Md. Faysal, Lecturer, CSE, UITS

Design and Implementation of a Blood Donation and Distribution Platform

Course Outcome

No.	COs
CO1	**Design** software systems to collect specifications and analyze project requirements.
CO2	**Ability** to use the techniques, skills, and modern engineering tools necessary for engineering practice.
CO3	**Develop** testing mechanisms for assuring software quality including the dependability and availability.
CO4	**Implement** real-life complex software engineering problems with teamwork problem solving skills
CO5	**Demonstrate** software system engineering management and economic decision making.

Problem Scenario

In many healthcare systems, there exists a critical gap between blood donors, hospitals, blood banks, and patients during emergency situations. Lack of a centralized system leads to delays in blood availability, inefficient communication, and poor inventory management.

To solve this issue, a **Blood Donation and Distribution Platform** is required that enables:

- donor registration and management
- hospital and blood bank coordination
- real-time blood inventory tracking

- emergency blood request handling
- location-based donor search
- notification and approval system

The system must ensure:

- high reliability
- fast response time
- secure data handling
- scalability
- cost-effectiveness

Students are required to design and implement this system as a **complete software engineering project**.

Project Objectives

Students will:

- analyze system requirements
- design software architecture
- select appropriate technologies
- implement a working system
- perform testing and validation
- manage project development process
- justify engineering decisions

Project Phases and Tasks with KPA Alignment

Phase	Activities	CO	K (Knowledge Profile)	P (CEP)	A (Engineering Activity)
Phase 1: Requirement Analysis	• Identify stakeholders (donors, hospitals, blood banks, patients) • Collect functional & non-functional requirements • Define system scope and constraints • Prepare SRS document	CO1	K2, K3	P3, P6	A2
Phase 2: System Modeling & Technology Selection	• Select SDLC model (Agile/Waterfall) • Choose technology stack (frontend, backend, database)	CO2	K2, K4, K6	P3, P4	A1, A3

Phase	Activities	CO	K (Knowledge Profile)	P (CEP)	A (Engineering Activity)
Phase 3: System Design	● Design use case diagrams & system flow ● Select tools, APIs, hosting platforms ● Design system architecture (3-tier/microservices) ● Design database schema (ER diagram) ● Develop UI/UX prototypes ● Define module interactions (authentication, inventory, requests)	CO1, CO2	K3, K5	P3, P7	A2, A3
Phase 4: Implementation	● Develop frontend interface ● Develop backend services and APIs ● Database integration ● Implement modules (donor, hospital, request, admin dashboard)	CO2, CO4	K4, K5, K6	P4, P7	A1, A2, A5
Phase 5: Testing & Maintenance	● Unit testing ● Integration testing ● System testing ● Debugging ● Performance & reliability testing ● Maintenance planning	CO3	K3, K4, K6	P6, P7	A2, A5
Phase 6: Project Management & Evaluation	● Task distribution ● Timeline & milestone tracking ● Cost estimation ● Resource management ● Final documentation and reporting	CO5	K5, K6, K7	P2, P6	A1, A5

Functional Requirements

The system must include:

1. User registration (donor, hospital, admin)
2. Blood group search system
3. Blood inventory management
4. Emergency blood request system
5. Notification and alert system
6. Request approval workflow
7. Location-based donor search
8. Admin dashboard
9. Report generation
10. Real-time status tracking

Non-Functional Requirements

- Security (authentication & authorization)
- Scalability (large number of users)
- Reliability (fault tolerance)
- Performance (fast response system)
- Usability (user-friendly interface)
- Maintainability (modular design)

Engineering Constraints

- Limited server and infrastructure resources
- Real-time emergency response requirement
- Data privacy and security concerns
- Network reliability issues
- Budget constraints
- System scalability requirements

Knowledge Profile (KPA Summary)

Category	Selected Codes
Knowledge Profile (K)	K2, K3, K4, K5, K6
Complex Engineering Problem (P)	P3, P4, P6, P7
Complex Engineering Activity (A)	A1, A2, A3, A5

Deliverables

Students must submit:

- Software Requirement Specification (SRS)
- System Design Document (UML + ER diagrams)
- Source Code (complete project)
- Testing Report
- Project Report
- Presentation Slides
- Zip all files of project and submission

Assessment Rubric

Criteria	Marks
Requirement Analysis	10
System Design	10
Implementation	30
Testing & Quality Assurance	10
Project Management & Documentation	20

Instructions for Students

1. Follow all project phases strictly.
2. Maintain proper documentation for each phase.
3. Use diagrams where necessary (UML, ER, architecture).
4. Ensure originality of work.
5. Show justification for all engineering decisions.
6. Teamwork and collaboration are mandatory.

Expected Learning Outcomes

After completing this assignment, students will be able to:

- analyze real-world software engineering problems
- design scalable and maintainable systems
- apply modern software engineering tools and practices
- implement complete software systems
- perform testing and validation
- work effectively in teams
- justify engineering decisions based on performance, reliability, and cost